

# State of *CryptAI*

A nine-specialist audit of the repo, the deployment, the cost model, and the quant evidence — every claim file:line-verified on branch `fix/m1-foundation-bugs` @ `8cb897e`, 2026-08-24.

| Doc              | Method                                   | Verdict                                             |
|------------------|------------------------------------------|-----------------------------------------------------|
| 01 of 06 — Audit | 9 parallel audit agents · 325 tool calls | Product built · operations broken · fixable in days |

# The product is built. The operations broke it. Both are fixable.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                         |                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                |
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| <div>TESTS GREEN</div> <div>912</div> <div>687 backend · 99 frontend · 126 Rust — verify.sh passes in 38s</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <div>BACKEND</div> <div>SUSPENDED</div> <div>Render bandwidth overage 5.63/5 GB, since 2026-08-10</div> | <div>FRONTEND</div> <div>LIVE</div> <div>cryptai-app.vercel.app · Vercel Hobby</div> | <div>MONTHLY SPEND</div> <div>\$0</div> <div>every service on a free tier — blocked, not billed</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <div>SINCE LAST AUDIT</div> <div>43</div> <div>commits — the session pipeline went from unwired to wired</div> |
| <div>WHAT CHANGED SINCE THE AUG-2 AUDIT — THE GOOD NEWS</div> <p>The Aug-2 verdict was "three good products that never met." <b>That gap is closed in code:</b> the session pipeline (start → per-user synthesis → proposal → approve → re-validate at live price → execute → per-position monitor) is genuinely wired into the one production entrypoint, with per-session LLM billing feeding a fail-closed cap. The Rust engine ships inside the production Docker image. Confirmed bugs #1–4 from the old audit (quota bypass, admin fail-open, unfed risk tracker, Delta crash) are <b>fixed with traced call paths</b>.</p> |                                                                                                         |                                                                                      | <div>WHY USERS SEE A DEAD PRODUCT ANYWAY</div> <ul style="list-style-type: none"><li>• <b>None of the 43 commits is deployed</b> — Render suspended the workspace before they could ship.</li><li>• The suspension's root cause is only <b>half-fixed</b>: the API's market feed was gated, but the daemon's DataAgent still opens ~2.2 GB/day of streams at startup (slide 4).</li><li>• Supabase has been idle 14 days — past its 7-day free-tier pause threshold; it likely needs a dashboard restore too.</li><li>• A newly found bug means the signal plane <b>cannot simply be switched on</b> (slide 5).</li></ul> |                                                                                                                |

# Subsystem truth table — construction-path verified, not test-verified

| SUBSYSTEM               | STATUS | EVIDENCE-BACKED REALITY                                                                                                                                        | CAVEAT THAT MATTERS                                                                                 |
|-------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Session pipeline        | WIRED  | Pool constructed in the daemon, daemon embedded in the API process; a metered session runs real analysis and creates proposals; clock pauses on propose        | —                                                                                                   |
| Per-user synthesis      | WIRED  | LLM selects the candidate fitting <i>this</i> trader + writes the thesis; never invents prices; forecast-language lint; deterministic fallback                 | Candidate <i>generation</i> still shared per symbol — selection is per-user                         |
| Approve → execute       | WIRED  | Re-validates expiry/invalidation/price-drift at live price, books through the user's risk gate; lost-reply recovery; 30s sweeper repairs orphans               | —                                                                                                   |
| Position monitors       | WIRED  | Supervisor unconditional; 1 unmetered monitor per position: time stop, profit give-back, adverse-pulse exit, reduce-only close                                 | <b>Do not survive a daemon restart</b> — paper engines never rehydrate; Render restarts are routine |
| Session LLM billing     | WIRED  | Usage normalized + booked even on failure paths, integer micro-USD, cap ends the session fail-closed                                                           | All-:free rotation prices at \$0 — the time quota is the only binding meter today                   |
| Risk limits             | FIXED  | PortfolioStateTracker fed by both production booking paths — free-tier caps now evaluate a real portfolio                                                      | Live (non-paper) close hook still missing — errs over-strict, the safe direction                    |
| Rust signal engine      | DARK   | Built into the production image, supervised child-process wiring complete, 126 tests                                                                           | Never enabled: <code>RUN_SIGNAL_ENGINE_IN_API</code> unset — and see the Pulse bug (slide 5)        |
| Hot-window surface      | ABSENT | No global pulse, no tradability history: <code>pulse_snapshots</code> has a table, indexes, a parity test — and <b>zero writers</b>                            | The founder's "hottest hour" has no data source until the writer ships                              |
| Tier/quota model        | GAP    | Every tier gets the same pooled 30 min/day; session count unlimited; channels not tier-gated; nothing blocks a new session while a trade is open               | The 2x30-min slot model + focus lock is R2 work (doc 03/06)                                         |
| Legacy single-bot stack | LEGACY | <code>execution_agent</code> / <code>order_tracker</code> / old monitor: tested but reachable only from <code>src/main.py</code> , which production never runs | Marked legacy; candidates for a later attic pass                                                    |

# The bandwidth post-mortem: streams that served nobody

Render free allows 5 GB/month of bandwidth — ~167 MB/day. On Aug 10 the workspace hit 5.63 GB, of which **5.61 GB was service-initiated** (17 MB actually served users). Two always-on Binance consumers did it — kept alive 24/7 by our own keepalive cron.

| CONSUMER                                                                                                                        | COST        | STATE ON HEAD                                                                                                                | FIX                                                               |
|---------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Daemon DataAgent</b> — depth20@100ms + markprice@1s + ticker + 5 kline TFs per symbol, subscribed unconditionally at startup | ~2.2 GB/day | <b>STILL IN CODE</b> — only the LLM analysis was demand-gated, never these streams. Resume today → re-suspended in ~2–3 days | Gate to scan demand + REST snapshots ( <b>R0, before resume</b> ) |
| <b>API base market feed</b> — btcusdt ticker + depth5@100ms + kline_1m, formerly subscribed at boot streaming to nobody         | ~216 MB/day | <b>FIXED</b> — refcounted to live dashboard viewers, 30s linger (commit 9edada0); idle backend subscribes nothing            | Deployed on resume                                                |
| <b>Engine bookTicker stream</b> — pushed on every best-bid/ask change                                                           | ~200 MB/day | <b>FIXED</b> — removed from the stream URL with a regression test; replaced by one batched REST poll per 5s                  | Deployed when engine ships                                        |
| <b>Engine kline WS</b> — 12 streams at 3 symbols × 4 TFs                                                                        | ~245 MB/day | <b>OVER BUDGET IF ENABLED</b> — 7.4 GB/mo alone                                                                              | REST-poll closed bars (R1) → ~2 MB/day                            |
| <b>Idle-socket churn</b> — zero-stream socket trips its own 90s staleness check, reconnects forever                             | ~6 MB/day   | <b>NEW FINDING</b> — ~960 pointless TLS handshakes/day + log noise                                                           | Skip staleness when no subscriptions (R0)                         |

IF RESUMED AS-IS

**~2.4** GB/day

14× budget — suspension #2 within days

AFTER R0 + R1 FIXES

**~1.5** GB/mo

viewer-gated feed + REST polling — 3.5 GB monthly headroom

THE LAW GOING FORWARD

**Name your consumer**

no stream or loop without a live session, viewer, or open position attached

# The defect list that matters now

| # | DEFECT                                                                                                             | SEVERITY | CONSEQUENCE IF SHIPPED                                                                                                            | WHEN            |
|---|--------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1 | Daemon DataAgent always-on streams                                                                                 | BLOCKER  | Re-suspension within days of resuming — product dies again                                                                        | R0 · pre-resume |
| 2 | Pulse dataclass vs <code>.get()</code> — new find, reproduced                                                      | HIGH     | Enabling the signal plane fails <b>every metered cycle</b> while the user's clock drains; tests pass only because they fake dicts | R1 · pre-enable |
| 3 | Monitors + paper desks lost on restart — engines never rehydrate                                                   | HIGH     | "Your agent never sleeps" is false after any deploy/restart — the product's core promise breaks silently                          | R2              |
| 4 | Symbol-universe mismatch — analysis runs BTC/ETH/XAUT, sessions run BTC/ETH/SOL                                    | MED      | ~1/3 of all LLM spend produces setups no session can use; SOL sessions scan a symbol with no candidates. One-line env fix         | R1 · env        |
| 5 | Quota model gaps — same 30 min/day for every tier, unlimited session count, no focus lock, channels not tier-gated | MED      | The tier ladder is unenforceable; free users can start sessions against an open trade and burn clock on undeliverable setups      | R2              |
| 6 | NEXT_PUBLIC_API_TOKEN — admin-capable static token labeled "read-only", plumbed as a frontend build arg            | MED      | One operator mistake ships an admin token in the public JS bundle                                                                 | R3              |
| 7 | Fail-open anonymous mode — no SUPABASE_URL + no API_AUTH_TOKEN ⇒ everyone sees all tenants' trades, and is admin   | MED      | Tenancy rests entirely on env vars being present                                                                                  | R3              |
| 8 | WS JWT in <code>?token=</code> + public <code>/metrics</code>                                                      | LOW      | JWTs land in proxy access logs; internals exposed                                                                                 | R3              |
| 9 | 188 quarantined tests (~21% of the suite)                                                                          | LOW      | Green pytest overstates coverage; precedent shows some quarantined tests encode real regressions                                  | ongoing         |

Fixed & verified since Aug 2 (for the record): session-quota clamp · admin fail-open for end-users · risk tracker fed by real bookings · Delta OrderStatus crash · session LLM billing. The uncommitted working-tree diff (analysis demand-gate completion) is **correct and worth keeping** — commit it after fixing its stale docstring and aligning `CYCLE_INTERVAL=180`.

# Dead weight is out — reference-checked, attic'd, gate still green

ONE-OFF SCRIPTS RETIRED

44

incl. nuclear\_clear\_positions.py & friends — **out of the production image**; scripts/admin.py is the guarded path

DEAD SRC MODULES

6

zero importers from any entrypoint or test — incl. EventCoordinator

SUPERSEDED DOCS

35+

Jan-2026 doc generation + 12-file fix-summary  
citation island

GATE AFTER THE MOVE

GREEN

verify.sh: typecheck ✓ 99 frontend ✓ import  
smoke ✓ 687 backend ✓

ABOUT THAT "300-SECOND ALWAYS-RUNNING LOOP"

The founder remembered an unused 300s loop. Found it: `EventCoordinator` — a 300s cycle trigger + 30s monitor. The twist: **it was never constructed anywhere**. It burned zero cycles; it was pure reading noise. The loop that actually cost money was the DataAgent's stream fleet (slide 4). Both are now handled: the corpse is in `_attic/src-dead/`, the live one is R0's blocker fix. **Lesson encoded in CLAUDE.md**: in this repo, "exists and passes tests" ≠ "runs" — always trace a constructor from an entrypoint.

WHERE EVERYTHING WENT

| LOCATION                                                                     | CONTENTS                                                                                                             |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <code>_attic/cta-scripts/</code>                                             | 44 one-off/destructive scripts (+ .dockerignore excludes _attic)                                                     |
| <code>_attic/src-dead/</code>                                                | event_coordinator, health_monitor, cycle_state, multi_context_analyzer, dual_context_llm_builder, ict_detector_CLEAN |
| <code>_attic/root-docs/</code> · <code>docs/</code> · <code>cta-docs/</code> | superseded doc generations + websocket test HTML + debug scripts                                                     |
| <code>_attic/README.md</code>                                                | what, why, and how to restore (git mv back)                                                                          |

Every move was reference-checked first; pointers in README, PRD, admin.py, data\_models.py, and eslint config were updated. Nothing deleted — git remembers, and the attic travels with the repo.

# LLM reality: dollars are \$0 — requests are the real currency

| CALL SITE (ALL → OPENROUTER :FREE ROTATION) | CADENCE                                        | PROMPT SIZE        | NOTE                                                                                                                         |
|---------------------------------------------|------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| Shared market analysis                      | per symbol per 180s cycle, demand-gated        | ~40k tokens        | ≈95% of all token throughput. Raw JSON dumps of market data — the #1 shrink target (send the engine's pulse summary instead) |
| Strategy creator                            | per symbol per cycle, when opportunities exist | ~1–2k              | Its purpose-built cache is constructed, tested, and <b>has zero callers</b> — the signature trap again; wire it or delete it |
| Per-session synthesis                       | once per session per cycle                     | ~0.7k in / 400 out | The only session-billed call. Per-user by design, correctly uncached                                                         |
| Trade monitoring                            | 15s polls                                      | 0                  | Fully deterministic — the "never sleeps" promise costs zero LLM                                                              |

THE FREE-CAP MATH (50 USERS × FREE TIER)

One 30-min session ≈ **40–70 requests / 0.7–1.3M tokens** (\$0 on :free). OpenRouter free caps ≈ 50 req/day — or **~1,000 req/day after a one-time \$10 credit** (verify current terms). 50 users staggered across the day ≈ 3,400 req/day = dead key. **Clustered into hot windows ≈ 560 req/day = it fits.** The hot-window mechanic isn't just UX — it is the cost model.

ORDERED COST LEVERS

- **Buy the one-time \$10 OpenRouter credit** — 20× the request cap; still effectively \$0/mo
- **Shrink the 40k-token analysis prompt** to a few k (pulse summary in, raw JSON out)
- **Fix the symbol mismatch** — recovers ~1/3 of spend instantly (env one-liner)
- **Hot-window clustering** — sessions share the demand-gated analysis cycles
- **Wire the strategy cache** (built, unwired) + keep analysis cache ≥2 cycles
- **Global daily request counter, fail-closed** — nothing tracks requests today; a dead-primary rotation day multiplies the 40k prompt by every retry
- Honest floor at scale: ~\$10–20/mo of paid inference once >10–15 staggered sessions/day — a Lite-tier pricing input, not a launch cost

# Quant verdict: a real filter, no directional edge, log everything

## ● VALIDATED — BUILD ON IT

- **Tradability = a NOW-cast filter** for move size over the next ~4–12h (IC +0.083/+0.096 at 4×1h bars; survives vol-normalization; decays to nothing by 24 bars).
- **Hour-of-day is the strongest factor** (residual IC +0.093, t +7.07) and is deterministic from the clock — the only honestly *forward* piece. London–NY overlap ≈ **17:30–20:30 IST**: perfect copy for Indian evening users.
- **Intraday is the only honest channel**: scalp fails validation (weights flip at 12 bars) *and* TDS economics; swing IC is insignificant at 24 bars, negative at 72.

## ● REFUTED / UNPROVEN — NEVER CLAIM

- **Regime→direction is dead** (walk-forward  $\Delta$ IC −0.008, a coin flip). The engine enforces this on the wire; the ban must now travel **up the stack** — the LLM synthesis layer is the only source of direction and carries **zero empirical validation**. Mirror the Rust contract test in the frontend.
- **No variant shows skill after multiple-testing correction** (deflated Sharpe 0.00 everywhere); no costs/slippage/funding modeled; 2 of 6 factors unmeasured.
- Therefore: **real-money launch is indefensible**. Paper-first is not caution — it's the only position the data permits.

## ● DAY-ONE LOGGING (FALSIFIABILITY)

- `pulse_snapshots` writer on every publish — full factor JSON + reference price (**zero writers today**; everything downstream is blocked on this).
- Forward-return backfill (15m/1h/4h + MFE/MAE) → re-run the walk-forward at +6 weeks; if temporal IC dies, the badge comes down.
- Per-proposal log: pulse id → channel → LLM direction → user decision → paper outcome — builds the direction-layer evidence that doesn't exist.
- Hour-of-day |return| buckets per symbol → replaces the hand-set liquidity map with learned hot hours.

## THE HONEST PRODUCT SENTENCE

"CryptAI ships a **validated market-conditions filter with live public calibration**, per-user trade construction, and unmetered position protection — on paper desks first, with real money gated behind six weeks of live evidence and a compliance floor." Design consequence: intraday holds (4–12h) **outlive 30-minute sessions** — the monitor isn't a feature, it's the product's spine.